

Ethan Wu

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EDUCATION

Northwestern University

Incoming, Computer Science & Operations Research (IEMS)

Starting September 2026 – June 2030 (Anticipated)

Evanston, IL

Commonwealth School

High School

September 2022 – June 2026

Boston, MA

- *Relevant Courses:* Multivariable Calculus, Linear Algebra, Real Analysis, AP Calculus BC, AP Statistics, AP Computer Science A, Designing Programming Languages (CS 5).
- Built 2 websites, both used by **200 students & teachers**.

SKILLS

Languages: Python, C++, Mojo, Rust, Go, JavaScript

Tooling & Frameworks: PyTorch, Google OR-Tools, Jax/Flax, LLVM, LightGBM, Optuna

Knowledge/Domains: Data Structures & Algorithms, Multi-Threading, Optimizations, Clustering Algorithms, ML

WORK EXPERIENCE

Research Intern

MIT Jameel Clinic - References provided upon request

October 2024 – Present

Cambridge, MA

- Performed clustering analysis on complex public health data; co-authoring **2 papers** currently in progress.
- Identified novel method for find the best clustering from clusterings generated with different hyperparameters.
- Replicated & verified cancer prediction computational models in published papers.
- Co-designed a **custom featurizer** for a cancer prognostic machine learning model with an MIT PostDoc.
- Co-organized a **~60 student summer camp** about AI in healthcare in 2025 and 2026: [See LinkedIn Post](#)
- In charge of designing, planning, and leading the summer program's hackathon.
- Built infrastructure software, such as a software deployment map and a summer program application portal.

PROJECTS

Strata - Mojo, LAPACK/BLAS

<https://prefix.dev/channels/modular-community/packages/strata>

- Working on a pure Mojo machine learning library with Scikit-Learn ergonomics and 0 Python interop
- Leveraged strict traits, static polymorphism, and generic SIMD types across float/int bit-widths to optimize ML models (Linear/Logistic Regression, Random Forests, K-Means)
- Applied LAPACK/BLAS to handle linear algebra computation. [See LinkedIn Post](#)
- One of the first 50 libraries accepted into Modular's official Modular-Community index on prefix.dev

Kluster-Fudge (PyPI) - Python, JIT, GPU

<https://pypi.org/project/kluster-fudge/>

- A Python library for clustering categorical data using K-Modes with various distance metrics (**> 3000 downloads**).
- Implemented multi-threading, GPU acceleration, and JIT optimizations for better performance.

cox-hazards (Crates.io) - Rust

<https://crates.io/crates/cox-hazards/>

- Built Rust library for survival analysis using Cox Proportional Hazards algorithm (**~1000 downloads**).
- Implements Newton-Raphson and AdamW optimizers, with L1/L2 regularization for feature selection.

Kategorize (Crates.io) - Rust

<https://crates.io/crates/kategorize>

- Built Rust library for K-Modes & K-Prototypes with various distance metrics (**~950 downloads**).
- Multi-threaded and optimized with distance caching.

VEX Robotics Match Prediction Model

- Used JAX to implement ResMLP with Huber Loss to make match predictions **with 75% accuracy**.

OPEN SOURCE CONTRIBUTIONS

CubeCL (TracelAI)

<https://github.com/tracel-ai/cubecl>

- Contributed upstream bug fix to CubeCL, a multi-platform GPU compute language extension for Rust.